

Vraj Patel

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M.S. Computer Science candidate building scalable, cloud-native backend services and AI-integrated systems end-to-end from design through deployment and production monitoring. Strong in Python, TypeScript, C++, microservices, REST APIs, and AWS. IEEE-published researcher and solo hackathon winner; ready to contribute from day one.

EDUCATION

M.S. Computer Science · *California State University, Fullerton* · GPA: 3.75/4.00 Aug 2024 – May 2026

B.Tech. Information & Communication Technology · *Marwadi University, India* · GPA: 3.7/4.00 Jul 2019 – Apr 2023

EXPERIENCE

Teaching Associate & Teaching Assistant · *California State University, Fullerton* Aug 2025 – May 2026

- Taught **CPSC 131 (Data Structures)** and **CPSC 335 (Algorithm Engineering)** to 100+ students; automated grading pipelines with Python and Gradescope across 200+ weekly submissions.
- Held office hours and led recitation sessions on **Dynamic programming, recursion, graph algorithms, and asymptotic analysis**; supported 200+ student debugging sessions across the semester.

Research Assistant · *Yuan-Ze University, Taiwan* Jul 2023 – Feb 2024

- Designed and benchmarked unsupervised **LSTM-Autoencoder** models for ECG anomaly detection on the MIT-BIH dataset using **TensorFlow/Keras**; co-authored **IEEE** peer-reviewed paper presented at ICCE 2024, Las Vegas.
- Implemented TensorFlow/Keras pipelines for **ECG signal preprocessing** and unsupervised anomaly scoring, achieving competitive accuracy with zero labeled training data; research cited in follow-on cardiac deep learning studies.

Software Developer · *Lucent Innovation, Ahmedabad* Feb 2023 – Jul 2023

- Built backend services with **Python, Django REST Framework, and PostgreSQL** for e-commerce platforms; designed **REST APIs** with **JWT** auth, **RBAC**, and pagination, and integrated third-party payment and logistics providers.
- Tuned **MySQL** indexes and query plans, configured **CI/CD pipelines** with **GitHub Actions**, and drove code reviews within two-week **Agile** sprints.

Software Developer Intern · *RKIT Software Private Limited, Rajkot* May 2022 – Jul 2022

- Built enterprise web modules using **C#, ASP.NET Core MVC, and SQL Server**; wrote **optimized** stored procedures and indexed queries.

PROJECTS

Intelmaxxing - Off-Grid Career Intelligence Platform · **Citrus Hack XI** · **Best Solo Hack Winner**

Next.js 16 · TypeScript · Gemma 4 (Novita / Ollama) · ElevenLabs | [Intelmaxxing.tech](#) | [GitHub](#)

- Engineered a **multi-agent orchestration** platform with **5 parallel AI agents** and dual-provider LLM routing (cloud **Novita** + **local Ollama** auto-fallback), generating ranked company dossiers in 18–30s from **HN, GitHub, Reddit, and YC signals**.
- Designed a noir-themed evidence board UI with ElevenLabs voice briefings and auto-generated cold-email drafts with verified CEO/CTO contact resolution.

GitConnect - AI-Powered Developer Mentorship Platform · **HackDartmouth X** · **\$1,000 Novita AI Prize**

React · Vite · GitHub API · Gemini AI · Tailwind CSS · Vercel | [git-connect.co](#)

- Solo-built and deployed in under 24 hours (only solo developer to win a prize track); matches students with mentors by analyzing GitHub commit history and language proficiency via **Gemini AI**.

TUI - Terminal UI Library for AI Agents

TypeScript · Node.js · Open Source | [GitHub](#) | [npm](#)

- Published an open-source npm library of interactive **terminal UI** primitives (**spinners, prompts, tables**) purpose-built for **AI coding agents** and human-in-the-loop **CLI** workflows.

GreenLane - AI Sustainability Shopping Assistant

TypeScript · React · Chrome Extensions API · Node.js · Meta ExecuTorch (On-Device AI) | [GitHub](#)

- Built a **privacy-first** Chrome extension that intercepts Amazon pages in real time and scores sustainability via **on-device inference (Meta ExecuTorch)**, with zero external data transmission.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

Frameworks: React, Next.js, Node.js, Express.js, Django, Tailwind CSS, Vite

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, GitHub Actions, CI/CD, Vercel

Databases: PostgreSQL, MySQL, MongoDB

Tools: Git, Postman, VS Code, Claude Code, Cursor

PUBLICATION

I. Farady, V. Patel, C.-C. Kuo, C.-Y. Lin. "ECG Anomaly Detection with LSTM-Autoencoder for Heartbeat Analysis," Proc. IEEE ICCE 2024, Las Vegas. DOI: [10.1109/ICCE59016.2024.10444327](https://doi.org/10.1109/ICCE59016.2024.10444327) (cited in follow-on cardiac deep learning research)